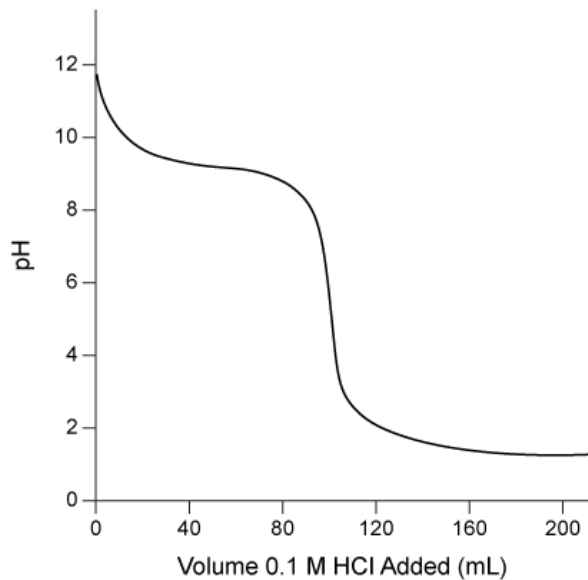


Consider the neutralization curve of aqueous ammonia with 0.1 M HCl, shown below.



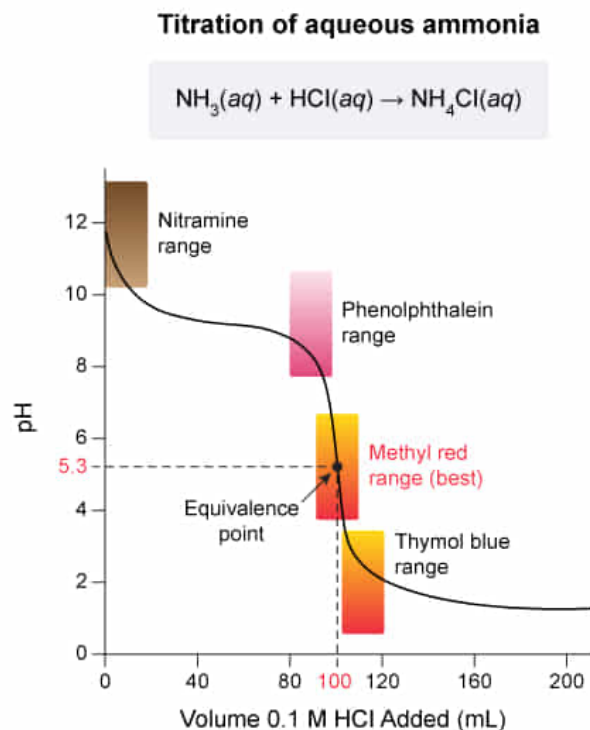
The best indicator to use in the titration is:

- ☐ A. thymol blue (changes color from pH 1.2 to 2.8). [9%]
- ✓ ☒ B. methyl red (changes color from pH 4.4 to 6.2). [78%]
- ☐ C. phenolphthalein (changes color from pH 8.2 to 10.0). [11%]
- ☐ D. nitramine (changes color from pH 10.8 to 13.0).

Correct

77% answered correctly

Explanation:



In an **acid-base titration**, a measured amount of an acid (or a base) solution (titrant) with a known concentration is slowly added to another solution containing an unknown concentration of a base (or an acid) to be measured. As the titration proceeds, an acid-base neutralization reaction occurs, producing a change in the pH of the solution being titrated.

When the acid (or base) being titrated is fully neutralized, the number of equivalents of acid exactly equals the number of equivalents of base (equivalence point). When passing through this point, the pH changes rapidly. On the titration curve, this is seen as a nearly vertical line, and at its midpoint lies the equivalence point.

To visually detect when a titration is complete, an **indicator** that has an **endpoint** (color change) near the pH of the **equivalence point** can be added to the solution. Different indicators **change color** across a particular **pH range** (endpoint range). The best indicator for a given titration is one that has a pH range that corresponds most closely to the pH of the equivalence point.

In the titration of aqueous ammonia, the equivalence point occurs at pH 5.3. Among the four indicators listed, methyl red has an endpoint range that overlaps closest to the equivalence point. Therefore, methyl red is the best indicator for the titration.

(Choice A) Thymol blue changes color after the equivalence point has already been reached. The endpoint of an indicator should occur closer to the equivalence point.

(Choice C) Phenolphthalein stops changing color well before the equivalence point is reached.

(Choice D) Nitramine only changes color under extremely basic conditions, and will not signal the equivalence point of this titration (at a weakly acidic pH).

Educational objective:

Indicators are added to a titrated solution during an acid-base titration to visually detect when a titration is complete. The best indicator for a titration is one that has an endpoint (color change) pH range nearest to the pH of the equivalence point.

Time spent:0 sec

Subject:

General Chemistry

QID:402031

System:

Solutions and Electrochemistry